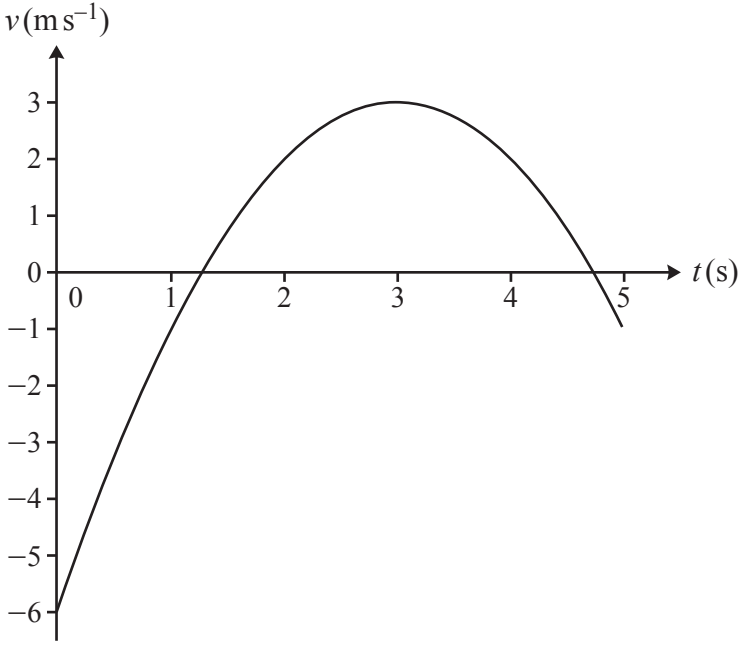
- The acceleration due to gravity is denoted by $g\,\mathrm{m\,s^{-2}}$. When a numerical value is needed use $g = 9.8$ unless a different value is specified in the question.

3 The velocity-time graph for the motion of a particle is shown below. The velocity $v\,\mathrm{m\,s^{-1}}$ at time $t\,\mathrm{s}$ is given by $v = -t^2 + 6t - 6$ where $0 \leqslant t \leqslant 5$.



(a) Find the times at which the velocity is $2\,\mathrm{m\,s^{-1}}$. **[2]**

(b) Write down the greatest speed of the particle. **[1]**